

Mathematical and Computational Methods

Exercise Sheet 5: Matrices

Tutorial route (►, in this order, about 80 minutes): 1.4, 2.2(a,b), 2.3(a,b), 2.4(a,b), 2.5(a,b), 2.7(a–c), 2.8(a–f). Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Matrix algebra

Exercise 1.1 (★ drill — addition, scalar multiple and product). Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 4 \\ -2 & 5 \end{pmatrix}$. Compute (a) $\mathbf{A} + \mathbf{B}$, (b) $2\mathbf{A} - \mathbf{B}$, (c) $\mathbf{AB}$.

Exercise 1.2 (★ drill — which products are defined).

(a) Matrices have sizes $\mathbf{A}: 2 \times 3$, $\mathbf{B}: 3 \times 3$, $\mathbf{C}: 3 \times 1$, $\mathbf{D}: 1 \times 2$. For each product $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{BC}$, $\mathbf{CD}$, $\mathbf{DA}$, $\mathbf{AC}$, state whether it is defined and, if so, its size.

(b) Compute $\mathbf{PQ}$ for $\mathbf{P} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix}$, stating its size first.

Exercise 1.3 (★ drill — transpose and trace). For $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$ write down $\mathbf{A}^T$ and compute $\text{tr}(\mathbf{A})$.

Determinants and inverses

Exercise 1.4 (► ★ drill — 2×2 determinants, then an inverse). Let $\mathbf{M} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 4 & 2 \\ 6 & 3 \end{pmatrix}$.

(a) Evaluate $\det \mathbf{M}$ and $\det \mathbf{N}$. Which matrix is invertible?

(b) Find the inverse of the invertible one and verify that its product with the original matrix is $\mathbf{I}$.

Exercise 1.5 (★ drill — cofactor expansion). Evaluate $\det \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}$ by cofactor expansion along the first row.

Linear systems

Exercise 1.6 (★ drill — solve by the inverse). Solve by the inverse method: $3x + 5y = 1$, $x + 2y = 1$. (You may reuse $\mathbf{M}^{-1}$ from Exercise 1.4.)

Exercise 1.7 (★ drill — solve by elimination). Solve by elimination (write the augmented matrix and reduce): $x + y = 5$, $2x - y = 1$.

Transformations

Exercise 1.8 (★ drill — *transformation matrices*). Write down the 2×2 matrices for (a) a rotation by 90° anticlockwise, (b) reflection in the x -axis, (c) an enlargement by factor 3. Then apply the rotation to $\mathbf{v} = (2, 1)$.

2 Tutorial problems

Exercise 2.1 (★★ core — $\mathbf{AB} \neq \mathbf{BA}$). Let $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

- (a) Compute $\mathbf{AB}$ and $\mathbf{BA}$ and confirm they differ.
- (b) Verify $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ for these matrices.

Exercise 2.2 (► ★★ core — *data science: shop revenue*). A shop records units sold of three products on two days as the rows of $\mathbf{S} = \begin{pmatrix} 4 & 2 & 1 \\ 0 & 3 & 5 \end{pmatrix}$ (rows = days, columns = products). The unit prices are $\mathbf{p} = (3, 5, 2)^\top$.

- (a) Explain why $\mathbf{Sp}$ is defined and what its entries mean.
- (b) Compute the daily revenue vector $\mathbf{Sp}$.

Exercise 2.3 (► ★★ core — *the threaded system, part 1*). Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$.

- (a) Show $\det \mathbf{A} = 1$.
- (b) Find $\mathbf{A}^{-1}$ by the cofactor (adjugate) method and verify $\mathbf{AA}^{-1} = \mathbf{I}$.

We reuse this matrix in Exercise 2.4.

Exercise 2.4 (► ★★ core — *the threaded system, part 2: two roads*). Solve $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ from Exercise 2.3 and $\mathbf{b} = (6, 5, 11)^\top$, that is

$$\begin{aligned} x + 2y + 3z &= 6, \\ y + 4z &= 5, \\ 5x + 6y &= 11, \end{aligned}$$

twice: (a) by the inverse method, and (b) by Gaussian elimination on the augmented matrix. Confirm the two answers agree.

Exercise 2.5 (► ★★ core — *infinitely many solutions, then none*).

- (a) Solve, by elimination,

$$x + 2y - z = 3, \quad 2x + y + z = 3, \quad x - y + 2z = 0,$$

and write the solution set as the vector equation of a line (Unit 4). How does the echelon form reveal that the system has infinitely many solutions?

- (b) Change the right-hand side of the third equation from 0 to 1 and apply the same row operations. What does the echelon form tell you now?

Exercise 2.6 (★★core — *spot the case*). Apply elimination to

$$x + y + z = 1, \quad x + 2y + 3z = 2, \quad 2x + 3y + 4z = 5,$$

and explain what the echelon form tells you about the solution set.

Exercise 2.7 (► ★★core — *physics: rotate then reflect*). A force $\mathbf{F} = (2, 3)$ (in newtons) is first rotated 90° anticlockwise and then reflected in the x -axis.

- Find the single 2×2 matrix $\mathbf{M}$ representing “rotate, then reflect”.
- Compute the transformed force $\mathbf{MF}$.
- What single geometric transformation is $\mathbf{M}$?

Exercise 2.8 (► ★★core — *conceptual, true or false*). Decide whether each statement is true or false; give a one-line reason or counterexample, and for (e) a proof.

- Matrix multiplication is commutative.
- Every square matrix has an inverse.
- $(\mathbf{AB})^\top = \mathbf{A}^\top \mathbf{B}^\top$.
- If $\det \mathbf{A} = 0$ then $\mathbf{Ax} = \mathbf{b}$ cannot have a unique solution.
- For all 2×2 matrices, $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$. (Prove it with $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.)
- There is a 2×2 matrix $\mathbf{A}$ with $\mathbf{Ax} = \mathbf{x} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ for every $\mathbf{x} \in \mathbb{R}^2$.

3 Further practice (home)

Matrix algebra

Exercise 3.1 (★★core — *data science: Markov chains (foreshadows Unit 6)*). A simple weather model has two states, *sunny* (S) and *rainy* (R). The *transition matrix* (columns = today, rows = tomorrow)

$$\mathbf{P} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$$

reads: if today is sunny, tomorrow is sunny with probability 0.8 and rainy with 0.2 (top and bottom of the first column), and similarly the second column handles a rainy today. Today is sunny, so the state vector is $\mathbf{x}_0 = (1, 0)^\top$.

- Compute $\mathbf{x}_1 = \mathbf{Px}_0$ and say what its entries mean.
- Compute $\mathbf{P}^2$, then $\mathbf{x}_2 = \mathbf{P}^2 \mathbf{x}_0$. Interpret the $(1, 1)$ entry of $\mathbf{P}^2$ (the probability of being sunny two days after a sunny day).

Determinants and inverses

Exercise 3.2 (★★core — *when is it singular?*). For which value(s) of k is each matrix singular?

$$(a) \begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}, \quad (b) \begin{pmatrix} 1 & 2 & 3 \\ 0 & k & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Exercise 3.3 (★★core — *inverse of a triangular matrix, then a system*). Let $\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

- Find $\mathbf{U}^{-1}$ by row-reducing $(\mathbf{U} \mid \mathbf{I})$, and verify $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$.
- Hence solve $x + y + z = 4$, $y + z = 2$, $z = 1$ by the inverse method.

Linear systems

Exercise 3.4 (★★core — *Gaussian, then Gauss–Jordan*). Consider $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

- Solve by Gaussian elimination (echelon form, then back-substitution).
- Continue the row operations from your echelon form to *reduced* row-echelon form (leading 1s, zeros above and below each pivot) and read the solution off directly.

Exercise 3.5 (★★core — *a one-parameter solution set*). Find all solutions of $x + y + z = 3$, $x - y + z = 1$ and write the answer as the vector equation of a line.

Exercise 3.6 (★★core — *physics: circuit currents*). Mesh analysis of a two-loop circuit gives the currents I_1, I_2 (in amperes) from

$$5I_1 - 2I_2 = 1, \quad -2I_1 + 4I_2 = 6.$$

Solve by the inverse method.

Transformations

Exercise 3.7 (★★core — *composing transformations, area and orientation*). Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (a horizontal stretch) and $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by 90°).

- Find the matrix for “stretch first, then rotate” and its determinant. Compute the other order $\mathbf{DR}$: does the order matter?
- On graph paper, sketch the unit square with corners $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$ and its images under $\mathbf{D}$ and under $\mathbf{RD}$. Find the area of each image and check that it equals $|\det|$ times the original area.
- Repeat (b) for $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. For each, compute the determinant, state the effect on area and orientation, and say whether the map is invertible.

Challenge / extension

Exercise 3.8 (★★★challenge — *inverse of a product*). Suppose $\mathbf{A}$ and $\mathbf{B}$ are invertible $n \times n$ matrices. Prove that $\mathbf{AB}$ is invertible with $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Exercise 3.9 (★★★challenge — *rotations add (links to Units 1 and 3)*). Let $\mathbf{R}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

- Explain geometrically why $\mathbf{R}(\theta)\mathbf{R}(\phi)$ must equal $\mathbf{R}(\theta + \phi)$.
- Compute $\mathbf{R}(\theta)\mathbf{R}(\phi)$ and, by comparing entries with $\mathbf{R}(\theta + \phi)$, recover the angle-addition formulae for $\cos$ and $\sin$ (stated in Unit 1). Compare with $e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}$ (Unit 3).

Exercise 3.10 (★★★challenge — matrices commuting with $\mathbf{J}$). Find all 2×2 matrices $\mathbf{X}$ that commute with $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, i.e. $\mathbf{XJ} = \mathbf{JX}$.